

BORRA BHAVITHA

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Experience

Engineering Intern

- **AI Research Intern at TANUH AI Center of Excellence, IISc (Present)** Contributing to an AI-driven healthcare project under the TANUH AI Center of Excellence
 - Developed and benchmarked deep learning models for mammography-based breast tissue density classification using PyTorch.
 - Built scalable training, evaluation, and preprocessing pipelines supporting supervised, self-supervised, semi-supervised, and domain adaptation techniques.
 - Designed and evaluated CNN- and Vision Transformer-based architectures, conducting extensive experimentation, hyperparameter optimization, and robustness analysis.
 - Improved cross-domain generalization through Domain-Adversarial Neural Networks (DANN), self-supervised learning, and explainable AI techniques.

Volunteer Roles

- **Workshop on Open Day, 2026** Demonstrated and explained the Tower of Hanoi problem during IISc Open Day, helping participants understand recursion through step-by-step visualization and interactive explanation.
- **Workshop on Open Day, 2025** Demonstrated an interactive facial expression recognition project, “Match the Emoji,” during IISc Open Day, where participants imitated displayed emojis and received real-time match scores generated by the model.
- **Pravega, IISc 2024** Volunteered for Pravega 2024 Culturals, serving as a point of contact for participants and supporting event coordination and smooth program execution.

Event Coordinator – Lasya (Culturals-Dance), Pravega, IISc 2025

- Organized and coordinated cultural events as part of IISc’s annual science and cultural fest.

Education

Bachelor of Technology (Mathematics and Computing) Indian Institute of Science (IISc) CGPA: 6.1	08/2024 – Present
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Class 11 & 12 (PCM) Narayana Junior College, Kanuru Grade: 98.3%	06/2022 – 06/2024
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Class 10 (CBSE), Kennedy High School Grade: 95.2%	06/2021 – 06/2022
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Skills

Programming: Python, C (basics), Bash

Core CS Concepts: Data Structures & Algorithms, Operating Systems , Recursion

Machine Learning: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Deep Learning, CNNs, Vision Transformers, Self-Supervised Learning, Semi-Supervised Learning, Domain Adaptation

Computer Vision: Feature Extraction, Classification, Data Augmentation

Systems & Optimization: Low-level Programming, Python-C Integration, Memory Management, Algorithm Optimization

Tools & Libraries: VS Code, NumPy, PyTorch, Scikit-learn, pydicom, Pandas, OpenCV, Git

Other: Problem Solving, Competitive Programming

Core Courses Completed

- Algorithms and Programming
- Analysis and Linear Algebra
- Computer Systems and Architecture
- Discrete Mathematics
- Linear Algebra and Multi Variable Calculus
- Numerical Methods
- Data Structures and Algorithms
- Probability and Statistics
- Introduction to Artificial Intelligence and Machine Learning
- Introduction to Algebraic Structures
- Introduction to Automata Theory and Computability
- Introduction to Basic Analysis

Projects

Self-Supervised Learning

- Representation Analysis (PyTorch) – Implemented SimCLR, MoCo, VICReg and Barlow Twins models. Analyzed robustness to label noise, representation quality and training dynamics using linear probing and embeddings.
- Observed improved generalization over supervised learning and sensitivity to early training corruption.

Neural Networks & Optimization from Scratch (NumPy / PyTorch)

- Implemented logistic regression and multi-layer neural networks from scratch with manual backpropagation and gradient-based optimization.
- Trained CNN models with validation-based hyperparameter tuning and implemented core evaluation metrics (precision, recall, F1, ROC).

Numerical Methods for Machine Learning Applications (Python)

- Implemented numerical algorithms (Newton, Modified Newton, Muller, interpolation, integration) and analyzed convergence, accuracy, and stability.
- Explored iterative and nonlinear systems through visualizations (Mandelbrot and Newton fractals) to understand optimization and convergence behavior.

Certifications

Introductory course on Quantum Computing

- One month course on Quantum Computing organized by Centre for Development of Advanced Computing, Hyderabad and Indian Institute of Technology, Roorkee with the support of Ministry of Electronics and Information Technology, Government of India.

Extra-curricular Activities

Tutoring

- Tutored Class 11 and 12 students preparing for JEE, conducting doubt clarification sessions and strengthening their understanding of core mathematical concepts and problem-solving techniques.

Event Coordinator – Telugu Samskrutika Samiti (TSS), IISc 2025 – Present

- Organized and conducted multiple cultural events for the Telugu student community.
- Coordinated volunteers, performers, and logistics to ensure smooth execution.
- Contributed to the decor team by designing and setting up stage and venue decorations to enhance the overall event experience.

Design Team

- Designed promotional posters and for tech and cultural events using Canva.

Dancer

- Classical (Learnt Kuchipudi for 5 years), Freestyle

Languages

English, Telugu, Hindi